

Nripesh

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Education

Vellore Institute of Technology (VIT)

B.Tech in Electrical and Electronics Engineering (CGPA: 8.46)

Vellore, Tamil Nadu

Sep 2022 – May 2026

Relevant Coursework: Power Systems Engineering, AC/DC Machines, Smart Grids, Control Systems, Industrial Automation, High Voltage Engineering, Power Electronics.

Professional Experience

NTPC Limited (Barh Super Thermal Power Station)

Barh, Bihar

Vocational Trainee / Electrical Maintenance & Operations

Jun 2025

- Analyzed operations of **660MW Supercritical Thermal units**, specifically Turbine-Generator synchronization and load dispatch.
- Monitored **Switchyard Operations (400kV/132kV)**, gaining practical exposure to Circuit Breakers, Isolators, and Transformers.
- Utilized **SCADA** systems to track real-time grid parameters, ensuring frequency stability.

Maven Silicon

Remote

Embedded System Design Intern

Jun 2024 – Jul 2024

- Developed an IoT-based control system using **Embedded C** to simulate industrial safety cut-offs.
- Integrated Temperature and GPS sensors to process telemetry data with low latency, optimizing system response time.

Key Projects

Intelligent PI Control Based Smart Bidirectional EV Charging & V2G Coordination

Jan 2026 – May 2026

- Developed a comprehensive **MATLAB/Simulink** model for a price-responsive, bidirectional EV charging system featuring a 100 kVA inverter and an LCL filter while strictly maintaining 20–90% battery SOC limits.
- Engineered a centralized **Energy Management System (EMS)** enabling peer-to-peer energy transfer across a shared DC bus during high-price peak hours, completely bypassing expensive grid power.
- Evaluated an ultra-local model-free **Intelligent PI (iPI) controller**, achieving **60% faster settling times** and 3x lower overshoot under dynamic pricing conditions while keeping grid current THD below 3% (IEEE 519).

Custom Water Leak Detector PCB

May 2026

- Engineered a custom hardware-based water leak detection circuit and printed circuit board from scratch using **KiCad 9.0**.
- Designed a reliable switching network utilizing an N-channel **2N7000 MOSFET** and a 10MΩ pull-down resistor to eliminate false triggering from ambient static electrical noise.

Vehicle-to-Grid (V2G) Bidirectional Power Flow Simulation

Aug 2025 – Dec 2025

- Simulated a bidirectional power flow system where EVs act as mobile energy storage to stabilize grid frequency during peak loads.
- Designed control logic in **MATLAB/Simulink** allowing EV batteries to store excess energy and feed it back seamlessly.

Smart Building Automation & Environment Control

Sep 2025

- Engineered an automated environment control system utilizing **ESP32** microcontrollers, PIR motion sensors, and LDR light sensors.
- Programmed logic in Embedded C++ for automatic load switching, reducing simulated energy consumption by 20%.

Real-Time Face Recognition and Notification System Using AWS

Nov 2024

- Developed a serverless face recognition pipeline leveraging **Amazon S3, Lambda, and Rekognition** for secure facial metrics verification.
- Configured real-time notification alerts distributed instantly via **Amazon SNS** upon detecting known faces.

Automotive Safety System

Jul 2024

- Designed an **Arduino-based** real-time tracking and emergency alert module utilizing GPS and GSM technologies.
- Engineered logic to instantly trigger and broadcast precise GPS coordinates to pre-registered contacts during critical fail-states.

Technical Skills

Electrical Core:	Power System Analysis, AC/DC Machines, Switchgear & Protection, PCB Design, Power Electronics.
Automation & Control:	SCADA, PID / iPI Controllers, PLC Logic, IoT Sensor Integration, BMS Concepts.
Software & Tools:	MATLAB, Simulink, Simscape, KiCad 9.0, AutoCAD (Electrical), LTSpice, AWS (S3, Lambda, Rekognition, SNS).
Hardware & Programming:	ESP32, Arduino, C/C++, Embedded C, Python.
Soft Skills:	Root Cause Analysis, Technical Troubleshooting, Industrial Safety, $\text{\LaTeX}$ Documentation.

Extracurricular & Achievements

Team RotorFPV (Drone Development Club)	VIT, Vellore
<i>Technical Member / Hardware Integration</i>	<i>Jul 2024 – Present</i>
– Calibrated Flight Controllers and tuned PID parameters ; managed precision soldering of BLDC motors and ESCs.	
• Winner , ISDC Drone Challenge, BITS Goa (2025).	
• 1st Runner-Up , Aero-Rush Technical Competition, NIT Trichy (2025).	
• 2nd Runner-Up , MultiRotor Competition, IIT Kanpur (2025).	